Assignment

Problem Description

The end of the semester is approaching, and Jimin suddenly discovers that there are still n assignments waiting to be finished.

For each assignment, Jimin already knows two things: how much time it takes to complete, and the latest time by which it must be submitted. Jimin can start working immediately at time 0, but can only work on one assignment at a time. Once an assignment is started, it must be finished before moving on to another assignment.

An assignment is considered successfully submitted if it is completed no later than its deadline. Otherwise, it cannot be counted.

Help Jimin determine the maximum number of assignments that can be completed before their deadlines.

Input Format

The first line contains one integer n, the number of assignments.

The next n lines each contain two integers t_i and d_i, describing one assignment:

- t_i: the time required to complete the assignment.
- d_i: the deadline by which the assignment must be completed.

Output Format

Print the maximum number of assignments that can be completed no later than their deadlines.

Constraints

- $1 \leq n < 150000$
- $1 \leq t_i < d_i < 2^{31}$

Example Test Case

Sample Input 1

```
4
2 4
3 8
4 7
5 10
```

Sample Output 1

```
3
```

Explaination

One optimal strategy is to complete the assignments whose required times are 2, 3, and 5. Their completion times are:

- Assignment with time 2: completed at time 2, before deadline 4.
- Assignment with time 3: completed at time 5, before deadline 8.
- Assignment with time 5: completed at time 10, exactly at its deadline 10.

Therefore, the maximum number of assignments that can be submitted on time is 3.